



A Note on Counting the Multiplicities of Elastic Surface Waves Using Weyl's Law

XIAOHUAN JIANG,^{1,2,3} SHAOQIAN HU,³ HAO XU,¹ and RONGTANG ZHANG¹

Abstract—Surface wave dispersion curves are widely used to constrain earth velocity structures and are important to compute theoretical synthetic seismograms with a mode-summation approach. While the computation of dispersion curves requires searching roots of nonlinear functions, some high-mode may be missed with improper choice of searching steps. The asymptotic distribution of eigenvalues of the elastic wave equation can be used as auxiliary information to design a sophisticated scheme to compute the surface wave dispersion curves. In this study, we show the Weyl's law, combined with the Liouville transformation, can be exploited to derive asymptotic eigenvalue counting functions of elastic surface waves in a horizontally stratified or radially heterogeneous medium. We also show the derived formulation according to the Weyl's law, in its simple case, agrees with previous studies. The derived asymptotic eigenvalue counting functions are validated by comparison with numerical results. This study demonstrates the Weyl's law can be used to derive eigenvalue counting functions of surface waves in elastic media, and it is also possible to be applied to more complex media.

Keywords: Surface wave oscillation, surface wave higher modes, theoretical seismology, asymptotic analysis.

1. Introduction

Weyl's law, independently conjectured by Sommerfeld (1910) and Lorentz (1910), and first proved by Hermann Weyl in a series of papers during 1911–1915 (Weyl, 1911), describes the asymptotic

distribution of eigenvalues of the Laplacian equation in a bounded domain. In more detail, the Weyl's law claims the leading order asymptotics of the number of eigenvalues of a Laplacian equation is determined by the volume of the computational domain only (de Hoop et al., 2017). There currently are various ways to prove the Weyl's law strictly (Weyl, 1911; Courant & Hilbert, 1931; Wet & Mandl, 1950; Ivrii, 2016; de Hoop et al., 2017). Weyl's law has many applications in various scientific areas, ranging from black body radiation problem to multiparticle quantum theory (reviews can be seen in Arendt et al. (2009) and Ivrii (2016)). Recently, Weyl's law of elastic surface waves in the Cartesian coordinate is discussed by de Hoop et al. (2017) and Zhai (2018). Specifically useful for this study, Wet and Mandl (1950) provides a formulation of eigenvalue counting functions for the Schrödinger equation in heterogeneous media using the Wentzel-Kramers-Brillouin-Jeffreys (WKBJ) approximation.

The related problem in seismology is calculating the theoretical dispersion curves of surface waves in a multilayered medium, which requires searching the roots of the dispersion (Dunkin, 1965) or secular equations (Chen, 1993). The dispersion curves of surface waves are popularly used to constrain the earth's structures and are also required to compute the synthetic seismograms using the mode-summation approach (Gilbert & MacDonald, 1960; Harkrider, 1964). However, Because the dispersion or secular equations are nonlinear, some roots may be missed with improper searching steps (Chen, 1993; Fan et al., 2007). Therefore, it is useful to estimate the asymptotic distribution of the eigenvalues to design some adaptive searching root-searching schemes (Fan et al., 2007; Wu & Chen, 2016; Pan et al., 2022). In this study, we consider two important cases which are practically useful in seismology, that

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00024-023-03389-6>.

¹ School of Civil Engineering and Architecture, Wuhan Polytechnic University, Wuhan 430023, China. E-mail: xhjiang@whpu.edu.cn; xuhao33011@163.com; peterzhang@whpu.edu.cn

² State Key Laboratory of Geodesy and Earth's Dynamics, Innovation Academy for Precision Measurement Science and Technology, CAS, Wuhan 430077, China.

³ Hubei Subsurface Multi-scale Imaging Laboratory, School of Geophysics and Geomatics, China University of Geosciences, Wuhan 430074, China. E-mail: hushaoqian@cug.edu.cn

is, the horizontally-stratified medium and the radially heterogeneous earth model.

In a horizontally-stratified medium, Fan et al. (2007) derived the eigenvalue counting function for the Rayleigh waves using high-frequency analysis of a specific form of propagator matrix. The derivation is with some assumption, which is validated by numerical experiments (Equation (21) in Fan et al. (2007)). Pan et al. (2022) provides the corresponding counting function formula for the Love waves without much explanation. The derived eigenvalue counting functions have been used by some subsequent studies to compute dispersion curves (Wu & Chen, 2016, 2022; Pan et al., 2022). In a radially heterogeneous spherical earth model, the ‘exact earth flattening transform’ is derived so that the spherical earth models can be transformed to the horizontally-stratified models (Bhattacharya, 1996; Arora et al., 1996; Bhattacharya, 2005; Hu & Zhu, 2023). The asymptotic properties of the surface wave higher modes of a spherical model have also been explored using the mode-ray duality (Sato & Lapwood, 1977; Ben-Menahem & Singh, 2000; Zhao & Dahlen, 1993) and the Langer approximation (Langer, 1937; Woodhouse, 1978; Cormier, 1980; Wu et al., 2022). In these studies, alternative to the original dispersion equation, some other equations need to be solved for the eigenvalues. The alternative equations are easier to solve compared with the original dispersion equation, but they are still nonlinear.

In this study, we show the Weyl’s law, combined with the Liouville transformation, can be exploited to derive the asymptotic eigenvalue counting function of elastic surface waves in more complex media, which may be difficult to derive using other methods. We first demonstrate that the eigenvalue counting equations proposed by Fan et al. (2007) and Pan et al. (2022) in the horizontally-stratified media are consistent with, but is only a special case of the Weyl’s law. We then show how to obtain the eigenvalue counting function to a radially heterogeneous spherical earth model. We also validate the derived eigenvalue counting functions by comparison with numerical results. The eigenvalue counting functions in this study can be used to design some adaptive scheme for calculating theoretical surface wave dispersion curves in a horizontal-stratified medium or

radially heterogeneous spherical earth. Using the Weyl’s law, the eigenvalue counting function can be obtained for problems in more complex media in higher dimensions.

2. Theory

Because the media concerned in this study is depth- or radius-dependent, it suffices to use the Weyl’s law in 1-D cases. In a 1-D heterogeneous medium, the general form of the equation we consider is (Wet & Mandl, 1950)

$$\frac{d^2\Psi}{dx^2} + (\lambda - q(x))\Psi = 0, \quad (1)$$

where $\Psi(x)$ is to be solved, $q(x)$ is dependent on x , and λ is independent on x .

The Weyl’s law claims the eigenvalue counting function in the 1-D case can be asymptotically expressed as (Wet & Mandl, 1950)

$$N(\lambda) = \frac{1}{\pi} \int (\lambda - q(x))^{\frac{1}{2}} dx. \quad (2)$$

Here, the term asymptotic means $\lambda \rightarrow +\infty$.

In the following, we show how to apply this theory to surface wave dispersion computation in horizontally-stratified and radially heterogeneous spherical media.

2.1. Weyl’s Law in a Horizontally-Stratified Medium

We consider a medium containing N layers over halfspace, and the i -th layer is characterized by $(\mu_i, \lambda_i, \rho_i)$. μ and λ are the Lamé’s parameters, which are related to shear and compressional wave velocities as $\rho\beta^2 = \mu$ and $\rho\alpha^2 = \lambda + 2\mu$. In this study, we assume the model parameters in each layer are vertically heterogeneous but are continuous with a second degree. Practically, the special case that each layer is homogeneous is most widely used up to date, because homogeneity is required by the propagator matrix method (Thomson, 1950; Haskell, 1953) and its modified versions (Dunkin, 1965; Buchen & Ben-Hador, 1996; Herrmann, 2013; Kennett & Kerry, 1979; Chen, 1993). However, some other methods, including the numerical integration method

(Takeuchi & Saito, 1972), the finite element method (Lysmer, 1970; Kausel & Roësset, 1981), the spectral collocation method (Denolle et al., 2012), and the spectral element method (Treyssede, 2016; Hawkins, 2018; Shi et al., 2022), have been used to compute the dispersions of a horizontally-stratified medium, and they allow the model parameters to be vertically heterogeneous within each layer.

In the frequency domain, using Lamé's theorem (Aki & Richards, 1980; Ben-Menahem & Singh, 2000), the wave equation related to the Love wave in i -th layer is

$$\frac{d}{dz} \left(\mu_i \frac{dV}{dz} \right) + (\rho_i \omega^2 - \mu_i k^2) V = 0, \quad (3)$$

where k is the horizontal wavenumber, ω is the angular frequency, $V(x)$ is the depth-dependent SH-component displacement.

Using the Liouville transformation (Ben-Menahem and Singh, 2000, Page 108) $V(z) = F(z)/\sqrt{\mu_i}$, the wave equation for $V(x)$ is reduced to

$$\frac{d^2 F}{dz^2} + \left[\frac{\omega^2}{\beta_i^2} - k^2 - \frac{\mu_i''}{2\mu_i} + \left(\frac{\mu_i'}{2\mu_i} \right)^2 \right] F = 0, \quad (4)$$

where μ' and μ'' are the 1-st and 2nd partial derivatives with respect to z , respectively, and $N(k; \omega)$ is the eigenvalue counting function at k for angular frequency smaller than ω .

Using Eqs. (1) and (2), the eigenvalue counting function for Love wave in each layer can be calculated as

$$N_i^L(k; \omega) = \frac{1}{\pi} \text{Re} \left[\int_{z_{i-1}}^{z_i} \sqrt{\frac{\omega^2}{\beta_i^2} - k^2 - \frac{\mu_i''}{2\mu_i} + \left(\frac{\mu_i'}{2\mu_i} \right)^2} dz \right]. \quad (5)$$

Asymptotically, the eigenvalue counting function of the whole model is the summation of N_i for all layers

$$N^L(k; \omega) = \frac{1}{\pi} \sum_{i=1}^N \text{Re} \left[\int_{z_{i-1}}^{z_i} \sqrt{\frac{\omega^2}{\beta_i^2} - k^2 - \frac{\mu_i''}{2\mu_i} + \left(\frac{\mu_i'}{2\mu_i} \right)^2} dz \right]. \quad (6)$$

Note that the eigenvalue counting function is always real.

The P-SV problem for Rayleigh wave is more complex. Here we directly use the results of de Hoop et al. (2017) and Zhai (2018) (Section (6.1) in

de Hoop et al. (2017)). The Rayleigh wave eigenvalue counting function can be calculated as

$$N^R(k; \omega) = \frac{1}{\pi} \sum_{i=1}^N \text{Re} \left[\int_{z_{i-1}}^{z_i} \sqrt{\frac{\omega^2}{\beta_i^2} - k^2 - \frac{\mu_i''}{2\mu_i} + \left(\frac{\mu_i'}{2\mu_i} \right)^2} dz \right] + \frac{1}{\pi} \sum_{i=1}^N \text{Re} \left[\int_{z_{i-1}}^{z_i} \sqrt{\frac{\omega^2}{\alpha_i^2} - k^2 - \frac{(\lambda_i + 2\mu_i)''}{2(\lambda_i + 2\mu_i)} + \left(\frac{(\lambda_i + 2\mu_i)'}{2(\lambda_i + 2\mu_i)} \right)^2} dz \right]. \quad (7)$$

2.2. Weyl's Law in a Radially Heterogeneous Spherical Earth

We consider a spherical model containing N -layer, and the i -th layer is characterized by $(\mu_i, \lambda_i, \rho_i)$. Similar as in the horizontally stratified medium, the model parameters in each layer may be vertically heterogeneous, but are continuous with the second degree. This requirement is useful since some global 1-D models, including the IASPEI91 model (Kennett & Engdahl, 1991) and the PREM model (Dziewonski et al., 1981) are described as polynomials in each layer.

The toroidal mode in the i -th layer of a radially heterogeneous earth model is governed by the equation (Ben-Menahem and Singh, 2000, Page 639)

$$\mu_i \left(\frac{d^2 V}{dr^2} + \frac{2}{r} \frac{dV}{dr} \right) + \frac{d\mu_i}{dr} \left(\frac{dV}{dr} - \frac{V}{r} \right) + \left[\rho_i \omega^2 - \frac{l(l+1)}{r^2} \mu_i \right] V = 0, \quad (8)$$

where r is the radius, and V is the toroidal displacement.

Using the Liouville transformation $V = \frac{rZ}{M_i}$ and $M_i = r^2 \sqrt{\mu_i}$. The wave equation can be written as (see Appendix A for detail)

$$\frac{d^2 Z}{dr^2} + \left(\frac{\omega^2}{\beta_i^2} - \frac{(l+1)(l-2)}{r^2} - \frac{M_i''}{M_i} \right) Z = 0. \quad (9)$$

Then the eigenvalue counting function of toroidal mode is

$$N^T(l; \omega) = \frac{1}{\pi} \sum_{i=1}^N \text{Re} \left[\int_{r_{i-1}}^{r_i} \sqrt{\frac{\omega^2}{\beta_i^2} - \frac{(l+1)(l-2)}{r^2} - \frac{M_i''}{M_i}} dr \right], \quad (10)$$

where M_i'' is the 2nd partial derivative of M_i with respect to r in the i -th layer.

Similar as for $P-SV$ problem in the horizontally-stratified medium, the eigenvalue counting function for the spheroidal mode is

$$N^S(l; \omega) = \frac{1}{\pi} \sum_{i=1}^N \operatorname{Re} \left[\int_{r_{i-1}}^{r_i} \sqrt{\frac{\omega^2}{\beta_i^2} - \frac{(l+1)(l-2)}{r^2} - \frac{M_i''}{M_i}} dr \right] + \frac{1}{\pi} \sum_{i=1}^N \operatorname{Re} \left[\int_{r_{i-1}}^{r_i} \sqrt{\frac{\omega^2}{\alpha_i^2} - \frac{(l+1)(l-2)}{r^2} - \frac{N_i''}{N_i}} dr \right], \quad (11)$$

where $N = r^2 \sqrt{\lambda + 2\mu}$, and N_i'' is the 2nd partial derivative of N with respect to r in the i -th layer.

3. Numerical Experiments

To verify the correctness of the eigenvalue counting functions using the Weyl's law to predict the multiplicities of surface wave dispersion. We consider several models in different scales, three horizontally-stratified models and the IASPEI91 earth model (Kennett & Engdahl, 1991). The synthetic dispersion curves are computed using the generalized reflection/transmission matrix method implemented by Wu and Chen (2022) for the horizontally-stratified media, and the spherical earth model is computed using the MINEOS software (Gilbert, 1971; Masters et al., 2011; Yang et al., 2010).

3.1. A Near-Surface Model

The first model is a near-surface model, which contains two layers over the halfspace (Table. 1) from Wu and Chen (2022). Both the dispersion curves of the Love (Fig. S1a) and Rayleigh waves (Fig. S1b)

Table 1
The parameters of Model 1

Layer	Thickness (m)	ρ (kg/m ³)	V_s (m/s)	V_p (m/s)
1	10	1800	100	240
2	5	1800	200	510
3	0	1800	300	800

are computed using the code of Wu and Chen (2022) in the frequency ranging from 0.01 to 300 Hz. There are a total of 2030 Love models and 6289 Rayleigh models whose eigenfrequencies are below 300 Hz. Then, the numbers of the eigenvalues could be counted below a certain phase velocity at each frequency (Fig. 1).

Then we used Eq. (6) to calculate the predictions of the numbers of the Love-wave eigenvalues (solid lines in Fig. 1a) at frequencies of 50 Hz, 100 Hz, 150 Hz, 200 Hz, and 300 Hz below certain phase velocities, respectively. The counting function in Eq. (6) predictions match well with the counting results (colored solid dots in Fig. 1a). A similar numerical test was used for Rayleigh-wave eigenvalue number counting with Eq. (7) (Fig. 1b). In the frequency range in our tests, the deviation of the counting numbers (defined as subtraction of numerical results from the predictions of the counting functions) for both the Love and Rayleigh waves are within ± 1 (Figs. 1c,d).

3.2. A Near-Surface Model with a Low-Velocity Layer

Previous studies show some modes may be trapped within the low-velocity layer of the model, and it is difficult to find these modes at certain high frequencies (Wu & Chen, 2016; Pan et al., 2022). In the second test, we test the performance of the eigenvalue counting functions using the near-surface model with a low velocity zone (Pan et al., 2022) as shown in Table 2. The Love and Rayleigh dispersion curves are shown in Fig. S2. The comparison between the results of the eigenvalue counting functions and the numerical results for Love and Rayleigh waves are shown in Figs. 2a and b. Results show the eigenvalue counting functions are still quite accurate (deviation within $-1 \sim 0.5$ for Love wave and $-1.5 \sim 1.0$ for Rayleigh waves, respectively) (Figs. 2c, d).

3.3. A Crustal Model

We next consider the Hadley-Kanamori velocity model, in which the Moho depth is 32 km (Model 3,

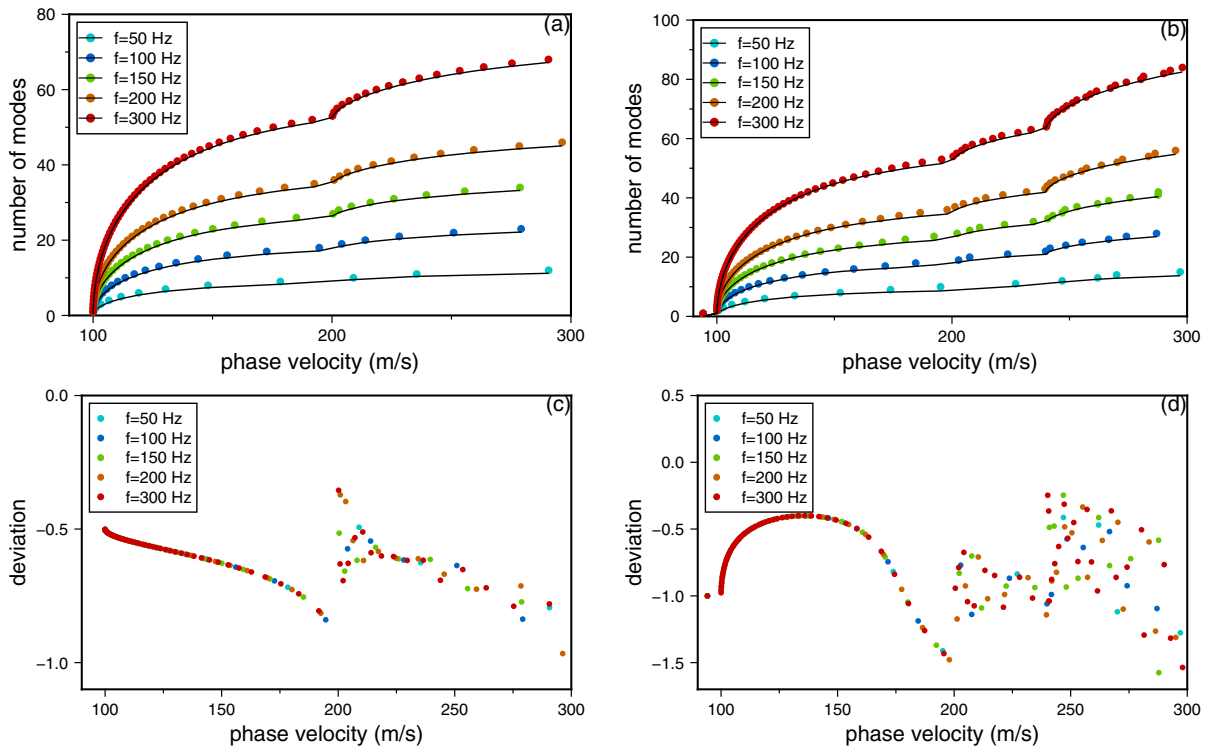


Fig. 1

The eigenvalue numbers of the Love-wave (a) and Rayleigh-wave (b) surface dispersion curves at different frequencies (50Hz, 100 Hz, 150 Hz, 200 Hz, and 300 Hz) below certain phase velocity for Model 1. Black solid lines are the counting function predictions, and colored solid dots represent numerical results. The deviations of the Love-wave (c) and Rayleigh-wave (d) eigenvalue numbers between the predictions of the counting functions and numerical results

Table 2

The parameters of Model 2

Layer	Thickness (m)	ρ (kg/m ³)	V_s (m/s)	V_p (m/s)
1	10	1780	190	1500
2	10	1850	350	1700
3	10	1800	240	1600
3	0	1940	600	1900

Table. 3) (Hadley & Kanamori, 1977). The Love-wave dispersion curves (Fig. S3a) and Rayleigh-wave dispersion curves (Fig. S3b) of this model were calculated in the frequency ranging from 0.01 to 4.0 Hz. Similar to the first test, the number of the modes of Love and Rayleigh waves were computed using Eq. (6) and Eq. (7). Comparisons between the numerical counting and the predictions of the counting functions below a certain phase velocity at frequencies of 1 Hz, 2 Hz, 3 Hz, 3.5 Hz, and 4 Hz

for Love wave (Fig. 3a) and Rayleigh wave (Fig. 3b) are shown. In our calculated frequency ranges, the results of counting functions have consistency with the numbers of modes using the numerical method for both the Love wave and Rayleigh wave (Figs. 3a,b). The deviation eigenvalue numbers for Love-wave and Rayleigh-wave dispersions are also mostly within ± 1 (Figs. 3c, d).

3.4. IASPEI91 Earth Model

Next, we test the counting function in Eqs. (10) and Eq. (11) using the IASPEI91 model. FigureS4 displays the eigenfrequencies of toroidal and spheroidal modes, respectively, as a function of angular degree l below 200 mHz for the IASPEI91 model. Different from the horizontally-stratified model, the numbers of eigenvalues were computed at each

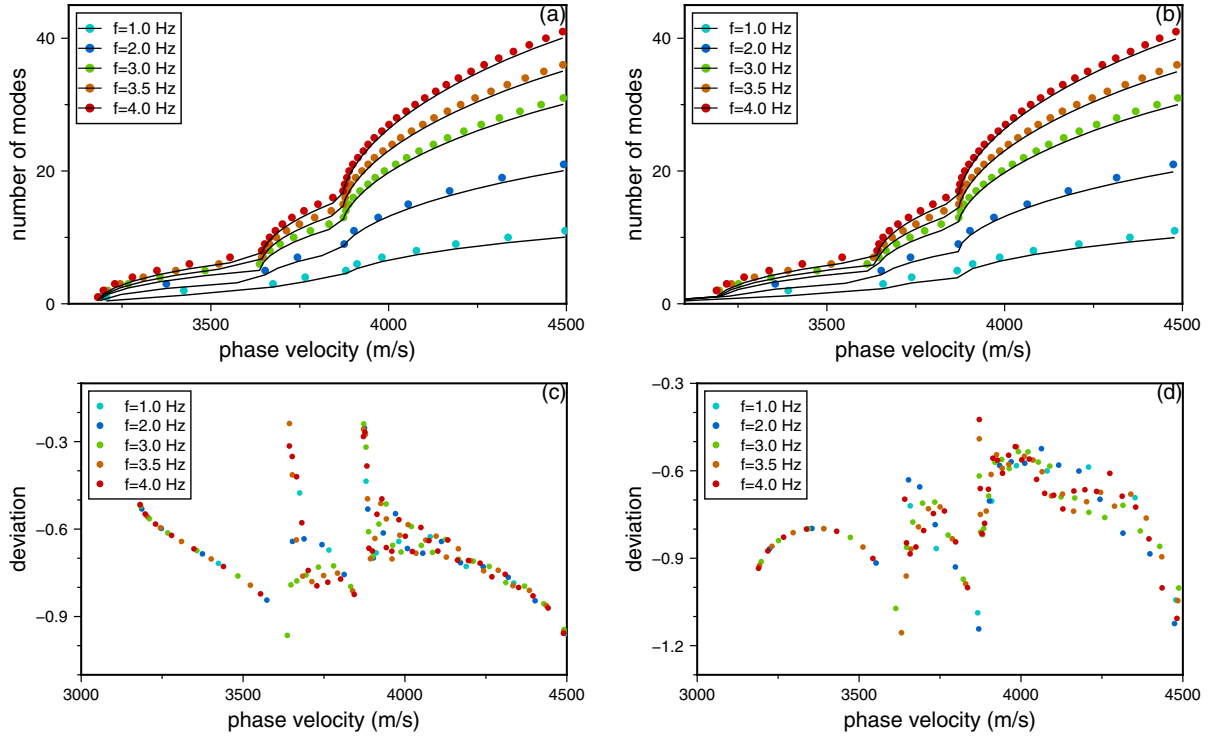


Fig. 2

The eigenvalue numbers of the Love-wave (a) and Rayleigh-wave (b) surface dispersion curves at different frequencies (20 Hz, 50 Hz, 80 Hz, 120 Hz, and 150 Hz) below certain phase velocity for Model 2. Black solid lines are the counting function predictions, and colored solid dots represent numerical results. The deviations of the Love-wave (c) and Rayleigh-wave (d) eigenvalue numbers between the predictions of the counting functions and numerical results

Table 3

The parameters of Model 3

Layer	Thickness (m)	ρ (kg/m^3)	V_s (m/s)	V_p (m/s)
1	5500	2530.4	3180	5501.4
2	10500	2786.3	3640	6300.8
3	16000	2913.7	3870	6699.0
4	0	3265.5	4500	7798.5

angular degree below certain frequencies. In detail, we counted the eigenvalues of the toroidal and spheroidal modes within given angular degree l ranges below 20, 100, 200, 500, and 1000, using Eqs. (10) and (11), respectively (solid black lines in Figs. 4a,b). Compared with the numerically counting results (colored solid dots in Figs. 4a,b), the errors of the predictions of the counting functions are less than 1 for toroidal modes (Fig. 4c) and less than 3 for spheroidal modes (Fig. 4d), respectively.

4. Discussion and Conclusion

Because the eigenvalue counting functions are expressed in the form of continuous integrals as in Eqs. (6), (7), (10), and (11). Numerical evaluation of the continuous integrals require discretization if the model parameter in a certain layer is heterogeneous. Also, note that there are extra terms containing parameter gradients in Eqs. (6), (7), (10), and (11) in a general case. When the models are homogeneously layered, the formulae will be degenerated to the case without these terms as in Fan et al. (2007); Pan et al. (2022), and the continuous integrals can be evaluated directly. Intuitively, the extra terms in the eigenvalue counting functions will be relatively more important for the models with a large velocity gradient. We evaluated the importance of the terms for the IAS-PEI91 model using numerical tests (Fig. 5). The numerical tests show the extra term is neglectable in

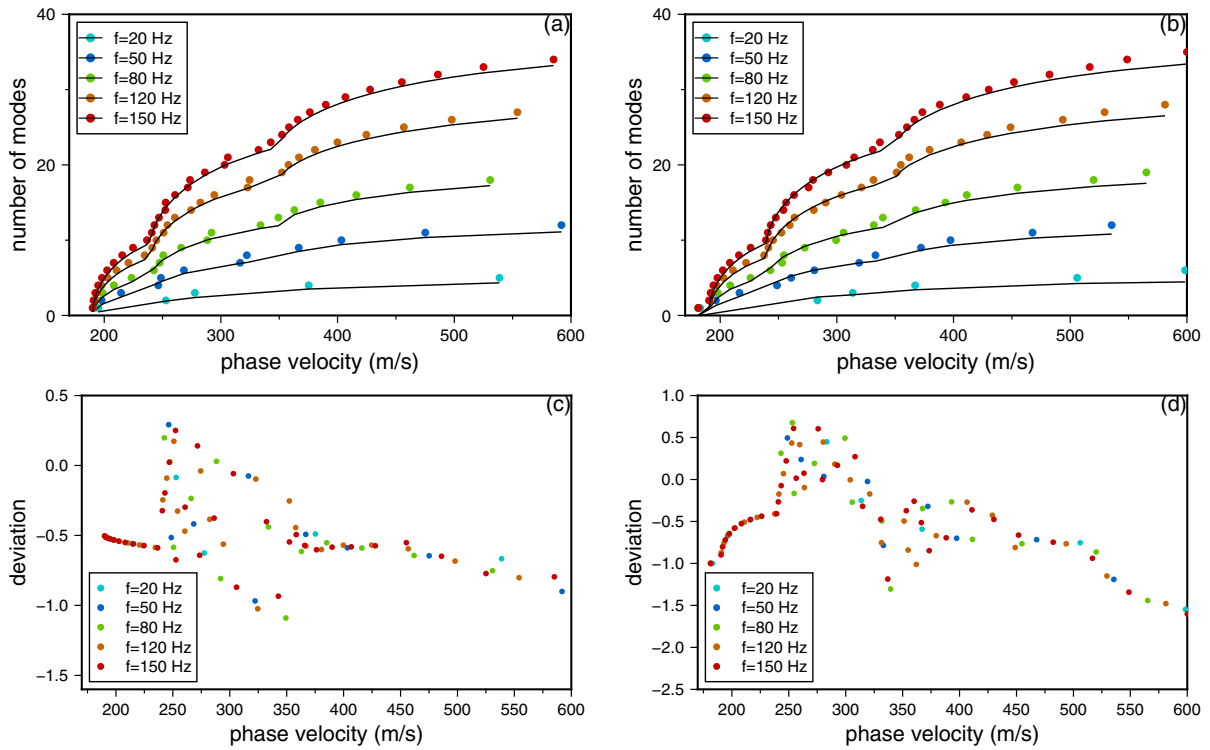


Fig. 3

The eigenvalue numbers of the Love-wave (a) and Rayleigh-wave (b) surface dispersion curves at different frequencies (1 Hz, 2 Hz, 3 Hz, 3.5 Hz, and 4 Hz) below certain phase velocities for Model 3. Solid lines are the counting function predictions and colored solid dots represent numerical results. The deviations of the Love-wave (c) and Rayleigh-wave (d) eigenvalue numbers between the predictions of the counting functions and numerical results

the frequency band we consider (Fig. 5). We then computed the effects of the extra terms in different models with different velocity gradients in the horizontally stratified media. Results show the terms can be safely neglected in most cases, and the errors introduced by the neglectation will be less than 0.05 at frequencies above 5 Hz (Fig. 6). It is also noteworthy the effects of the extra terms increases as the gradient increases at longer periods (Fig. 6). In these tests, we discretized the IASPEI91 model into 16900 thin layers, and divided the gradient layer of the horizontally-stratified medium into 100 thin layers, respectively.

Another issue to notice is that the earth models, including the IASPEI91 model, contain a liquid outer core. When computing the numbers of the eigenvalues of the toroidal modes, only the velocity models down to the upper bounds of the liquid outer core is required. The velocity models of the outer core and

inner core will not influence the values of the counting equations in Eq. (10). On the other hand, the shear wave velocity of the inner core will influence the number of modes of the spheroidal modes in Eq. (11).

In the horizontally-stratified medium, the total wave field can be represented as a sum of the normal modes and the leaky modes (Aki & Richards, 1980). Unlike the normal modes, the leaky mode attenuate exponentially with time because of leakage to the underlying halfspace, and they contain relatively weak energy (Aki & Richards, 1980). Because the Weyl's law used in this study is only valid in a bounded medium, it seems only the normal modes are included in the eigenvalue counting functions in Eqs. (7) and (6), and the numbers of the leaky mode are not included. However, It is also worthwhile to point out that, since the locked mode approximation can be used to compute the full synthetic

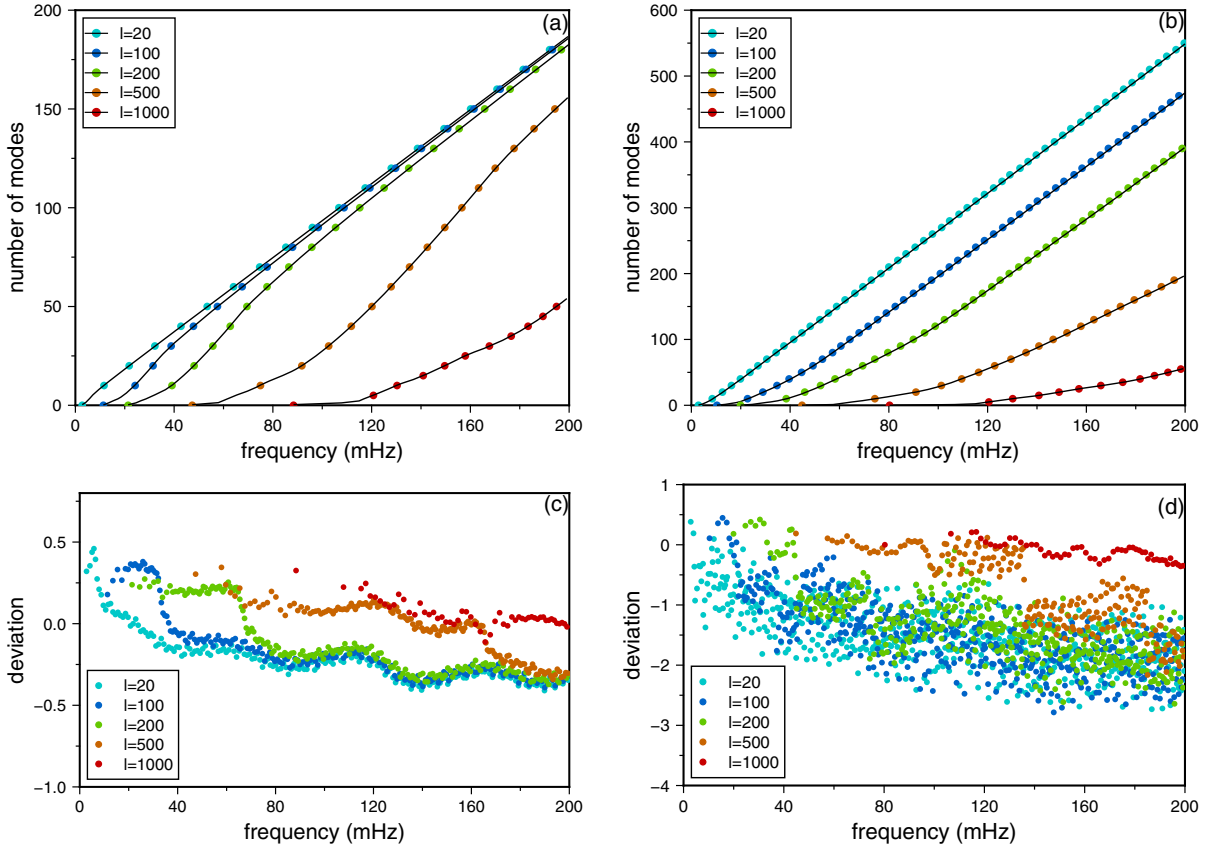


Fig. 4

The eigenvalue numbers of the toroidal (a) and spheroidal (b) surface dispersion curves within given angular degree l ranges below 20, 100, 200, 500, and 1000 for the IASPEI91 model. Solid lines are the eigenvalue counting function predictions and colored solid dots represent numerical results. The deviations of the toroidal and spheroidal eigenvalue numbers between the counting functions predictions and the numerical results are shown in c and d, respectively

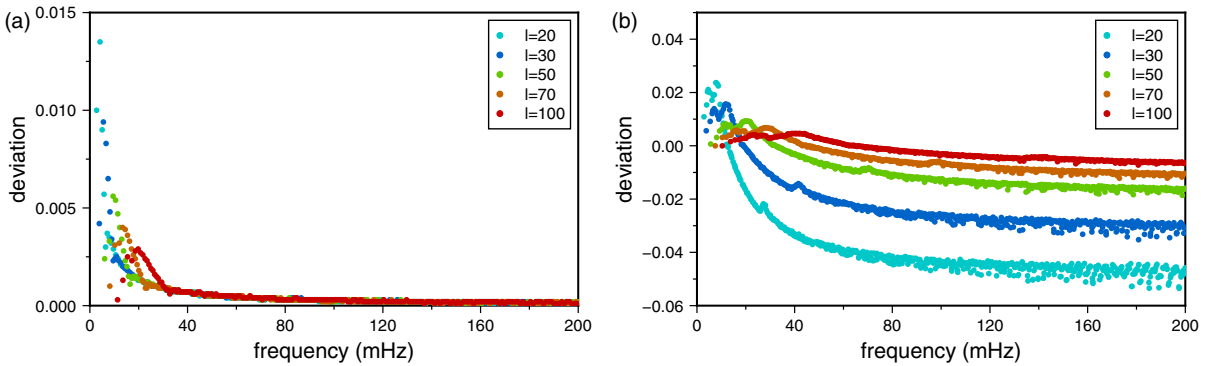


Fig. 5

The difference between the ‘exact’ eigenvalue counting functions and the ‘approximate’ eigenvalue counting functions without the extra terms containing velocity gradient for IASPEI91 model at different l ($l = 20, 30, 50, 70, 100$) below a certain frequency. a For toroidal-model; b For spheroidal-model

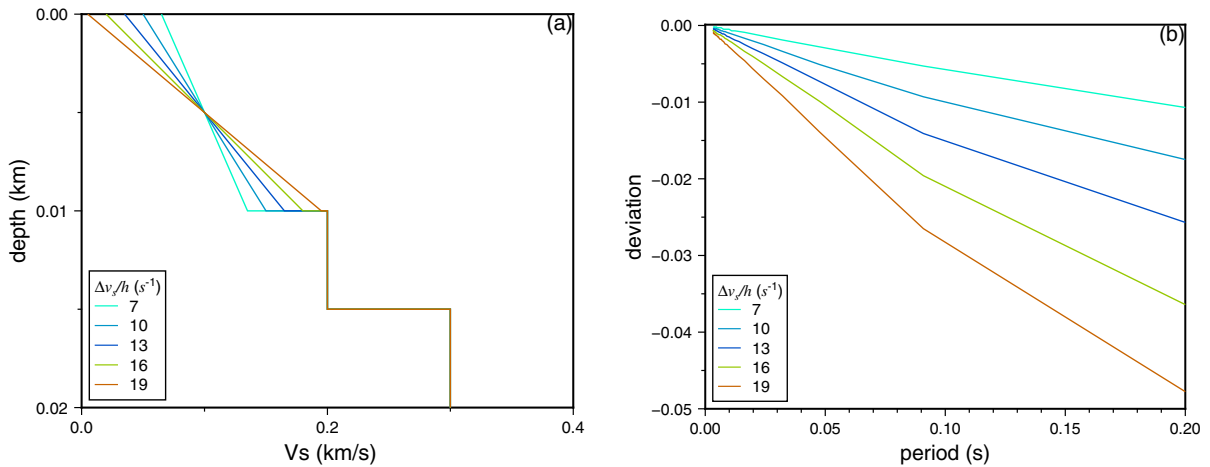


Fig. 6

a Horizontally stratified models modified from Model 1 (Table. 1). The modified models are with the velocity gradients (defined as $\Delta v_s/h$) of 7, 10, 13, 16, and 19 (s^{-1}) for V_s in their first layer. Their V_p/V_s ratio is fixed to be 2.4. **b** The difference between the ‘exact’ counting functions and the ‘approximate’ counting functions without the terms containing velocity gradient for different modified models in **a** at a certain frequency below 0.3 km/s for Rayleigh wave. Note that the x-axis in **b** is converted from frequency to period for visualization purpose

seismograms (Harvey, 1981), the locked mode approximation is also potentially applicable to study the properties of the leaky mode by adding a cap layer at the bottom of the model. In this study, we did not investigate the effectivity of the locked mode approximation to study leaky mode in detail.

In summary, in this study, we derive the asymptotic expression of the number of multiplicities using the Weyl's law. Numerical tests show that the formulae are quite accurate for the horizontally and radially stratified models. Tests also show that the extra term that accounts for the gradient of the model can be safely neglected in most applications unless the velocity gradient is dramatically large at low frequencies. Our results can be used to design a sophisticated scheme to search for roots for the surface wave dispersions.

Acknowledgements

Comments from two anonymous reviewers helped us to improve the manuscript. We are benefited from Prof. M. V. de Hoop's visit to USTC in the summer of 2018, and his introduction on the Weyl's law.

Author Contributions XJ and SH wrote the main manuscript text; HX and RZ review and edit the manuscript. All authors have read and agreed to the published version of the manuscript.

Funding

This study was supported by the National Natural Science Foundation of China (41904047), State Key Laboratory of Geodesy and Earth's Dynamics, Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences (SKLGED2022-4-2), Hubei Subsurface Multi-scale Imaging Key Laboratory (China University of Geosciences) (SMIL-2023-8), and Research Foundation of Wuhan Polytechnic University (2022RZ015).

Data Availability

There are no new data associated with this article.

Declarations

Conflict of interest The authors have not disclosed any competing interests.

Appendix A Derivation of the Liouville Transformation of the Toroidal Mode

Assuming $V(r) = rZ/M$, where Z is dependent on r , there are relations

$$\frac{dV}{dr} = \frac{r}{M}Z' + \left(\frac{1}{M} - \frac{rM'}{M^2}\right)Z, \quad (12)$$

and

$$\begin{aligned} \frac{d^2V}{dr^2} &= \frac{r}{M}Z'' + \left(\frac{2}{M} - \frac{2rM'}{M^2}\right)Z' \\ &+ \left(-\frac{2M'}{M^2} + \frac{2r}{M^3}(M')^2 - \frac{r}{M^2}M''\right)Z, \end{aligned} \quad (13)$$

where Z' and Z'' are the 1st and 2nd partial derivatives with respect to r , respectively, the meaning of other symbols can be seen in the main texts.

The wave equation for toroidal mode can be reorganized as

$$\mu V'' + \left(\frac{2\mu}{r} + \mu'\right)V' + \left(\rho\omega^2 - k^2\mu - \frac{\mu'}{r}\right)V = 0. \quad (14)$$

Substituting Eqs. (12) and (13) into Eq. (14), and forcing the term with Z' to be 0, we have

$$\mu\left(\frac{2}{M} - \frac{2rM'}{M^2}\right) + \left(\frac{2\mu}{r} + \mu'\right)\frac{r}{M} = 0. \quad (15)$$

Therefore,

$$\frac{M'}{M} = \frac{2}{r} + \frac{1}{2}\frac{\mu'}{\mu}. \quad (16)$$

Solving Eq. (16), we have $M = r^2\sqrt{\mu}$.

Combining Eqs. (12), (13), and (16), the wave equation related to Z can be simplified as

$$Z'' + \left(\frac{\omega^2}{\beta^2} - \frac{(l-1)(l+2)}{r^2} - \frac{M''}{M}\right)Z = 0 \quad (17)$$

Note there seems a typo in Ben-Menahem and Singh (2000) on page 643.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a

publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

REFERENCES

- Aki, K., & Richards, P. G. (1980). *Quantitative Seismology, Theory and Methods*. Freeman and Company.
- Arendt, W., Nittka, R., Peter, W., & Steiner, F. (2009). Weyl's law: spectral properties of the Laplacian in mathematics and physics. In: W. Arendt & W. P. Schleich (Eds.) *Mathematical Analysis of Evolution*. Information and Complexity. Wiley-VCH Verlag.
- Arora, S., Bhattacharya, S. N., & Gogna, M. L. (1996). Rayleigh wave dispersion equation for a layered spherical earth with exponential function solution in each shell. *Pure and Applied Geophysics*, 147, 515–536.
- Ben-Menahem, A., & Singh, S.J. (2000). *Seismic Waves and Sources*, 2nd, Corr. Ed. Springer
- Bhattacharya, S. N. (1996). Short notes: Earth-flattening transformation for $P-SV$ waves. *Bulletin of the Seismological Society of America*, 86, 1979–1982.
- Bhattacharya, S. N. (2005). Synthetic seismograms in a spherical Earth using exact flattening transformation. *Geophysical Research Letters*, 32(21), 303–306.
- Buchen, P. W., & Ben-Hador, R. (1996). Free-mode surface-wave computations. *Geophysical Journal International*, 124, 869–887.
- Chen, X. (1993). A systematic and efficient method of computing normal modes for multilayered half-space. *Geophysical Journal International*, 115, 391–409.
- Cormier, V. F. (1980). The synthesis of complete seismograms in an earth model specified by radially inhomogeneous layers. *Bulletin of the Seismological Society of America*, 70, 691–716.
- Courant, R., & Hilbert, D. (1931). *Methoden der Mathematischen Physik* (Vol. 1). Julius Springer.
- Denolle, M. A., Dunham, E. M., & Beroza, G. C. (2012). Solving the surface-wave eigenproblem with Chebyshev spectral collocation. *Bulletin of the Seismological Society of America*, 102(3), 1214–1223.
- Dunkin, J. W. (1965). Computation of modal solutions in layered elastic media at high frequencies. *Bulletin of the Seismological Society of America*, 55, 335–358.
- Dziewonski, A. M., Chou, T. A., & Woodhouse, J. (1981). Determination of earthquake source parameters from waveform data for studies of global and regional seismicity. *Journal of Geophysical Research*, 86(B4), 2825–2852.
- Fan, Y., Chen, X., Liu, X., Liu, J., & Chen, X. (2007). Approximate decomposition of the dispersion equation at high frequencies and the number of multimodes for Rayleigh waves. *Chinese Journal of Geophysics (in Chinese)*, 50(1), 233–239.
- Gilbert, F. (1971). Excitation of the normal modes of the Earth by earthquake sources. *Geophysical journal of the Royal Astronomical Society*, 22, 223–226.
- Gilbert, F., & MacDonald, G. J. F. (1960). Free oscillation of the Earth I. toroidal oscillations. *Journal of Geophysical Research*, 65(2), 675–693.

- Hadley, D., & Kanamori, H. (1977). Seismic structure of the Transverse Ranges, California. *Geological Society of America Bulletin*, 88, 1469–1478.
- Harkrider, D. G. (1964). Surface waves in multilayered elastic media I. Rayleigh and Love waves from buried sources in a multilayered elastic half-space. *Bulletin of the Seismological Society of America*, 54(2), 627–679.
- Harvey, D. J. (1981). Seismogram synthesis using normal mode superposition: The locked mode approximation. *Geophysical Journal of the Royal Astronomical Society*, 66, 37–69.
- Haskell, N. A. (1953). The dispersion of surface waves on multilayered media. *Bulletin of the Seismological Society of America*, 43, 17–34.
- Hawkins, R. (2018). A spectral element method for surface wave dispersion and adjoints. *Geophysical Journal International*, 215, 267–302.
- Herrmann, R. B. (2013). Computer programs in seismology: An evolving tool for instruction and research. *Seismological Research Letters*, 84, 1081–1088.
- Hoop, M., Iantchenko, A., Nakamura, G., & Zhai, J. (2017). Semiclassical analysis of elastic surface waves. arXiv preprint [arXiv:1709.06521](https://arxiv.org/abs/1709.06521).
- Hu, S., & Zhu, L. (2023). Computing theoretical seismograms from a point source in a spherical multilayered medium. *Seismological Research Letters*. <https://doi.org/10.1785/0220230173>
- Ivrii, V. (2016). 100 years of Weyl's law. *Bulletin of Mathematical Sciences*, 6, 379–452. <https://doi.org/10.1007/s13373-016-0089-y>
- Kausel, E., & Roësset, J. M. (1981). Stiffness matrices for layered soils. *Bulletin of the Seismological Society of America*, 71(6), 1743–1761.
- Kennett, B. L. N., & Engdahl, E. R. (1991). Travel times for global earthquake location and phase association. *Geophysical Journal International*, 105, 429–465.
- Kennett, B. L. N., & Kerry, N. J. (1979). Seismic waves in a stratified half space. *Geophysical Journal of the Royal Astronomical Society*, 57, 557–583.
- Langer, R. E. (1937). On connection formulae and the solution of the wave equation. *Physical Review*, 51, 669–676.
- Lorentz, H. A. (1910). Alte und neue Fragen der Physik. *Physikal. Zeitschr*, 11, 1234–1257.
- Lysmer, J. (1970). Lumped mass method for Rayleigh waves. *Bulletin of the Seismological Society of America*, 60(1), 89–104.
- Masters, G., Woodhouse, J.H., & Freeman, G. Mineos V1.0.2 [software]. Computational Infrastructure for Geodynamics. <https://geodynamics.org/cig>.
- Pan, L., Yuan, S., & Chen, X. (2022). Modified generalized R/T coefficient method for surface-wave dispersion-curve calculation in elastic and viscoelastic media. *Bulletin of the Seismological Society of America*, 112(5), 2280–2296.
- Sato, R., & Lapwood, E. R. (1977). The asymptotic distribution of torsional eigenfrequencies of a spherical shell. *I. J. Phys. Earth*, 25, 257–282.
- Shi, C., Ren, H., Li, Z., & Chen, X. (2022). Calculation of normal and leaky modes for horizontal stratified models based on a semi-analytical spectral element method. *Geophysical Journal International*, 230, 1928–1947.
- Sommerfeld, A. (1910). Die Greensche Funktion der Schwingungsgleichung für ein beliebiges Gebiet. *Physikal. Zeitschr*, 11, 1057–1066.
- Takeuchi, H., & Saito, M. (1972). Seismic surface waves. In B. A. Bolt (Ed.), *Methods in Computational Physics* (Vol. 11, pp. 217–295). Academic Press.
- Thomson, W. T. (1950). Transmission of elastic waves through a stratified solid medium. *Journal of Applied Physics*, 21, 89–93.
- Treysede, F. (2016). Spectral element computation of high-frequency leaky modes in three-dimensional solid waveguides. *Journal of Computational Physics*, 314, 341–354.
- Wet, J. S. D., & Mandl, F. (1950). On the asymptotic distribution of eigenvalues. *Proceedings for the Royal Society A*, 200, 572–580. <https://doi.org/10.1098/rspa.1950.0039>
- Weyl, H. (1911). Über die Asymptotische Verteilung der Eigenwerte. *Nachr. Konigl. Ges. Wiss. Göttingen*, 110–117.
- Woodhouse, J. H. (1978). Asymptotic results for elastodynamic propagator matrices in plane-stratified and spherically-stratified earth models. *Geophysical Journal of the Royal Astronomical Society*, 54, 263–280.
- Wu, B., & Chen, X. (2016). Stable, accurate and efficient computation of normal modes for horizontal stratified models. *Geophysical Journal International*, 206, 1281–1300.
- Wu, B., & Chen, X. (2022). A versatile solver of the normal modes for horizontal stratified complicated models. *Seismological Research Letters*, 93, 1852–1867. <https://doi.org/10.1785/0220210266>
- Wu, B., Zhao, L., & Chen, X. (2022). Uniformly asymptotic eigensolutions of the Earth's toroidal modes. *Geophysical Journal International*, 228, 250–258.
- Yang, H.-Y., Zhao, L., & Hung, S.-H. (2010). Synthetic seismograms by normal-model summation: a new derivation and numerical examples. *Geophysical Journal International*, 183(3), 1613–1632.
- Zhai, J. (2018). Analysis of inverse boundary value problems for elastic waves. PhD thesis, Rice University, Houston, Texas, USA
- Zhao, L., & Dahlen, F. A. (1993). Asymptotic eigenfrequencies of the Earth's normal modes. *Geophysical Journal International*, 115, 729–758.